VISHNU DHARSAN A S

Software Engineer | Data Analytics & GenAI

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PROFESSIONAL SUMMARY

Software Engineer with 1+ year of experience in data analytics, SQL, Python-based data cleansing, ETL pipelines, and Power BI dashboards, combined with hands-on GenAI experience (RAG, LangChain, Ollama). Skilled at turning raw data from multiple sources into reliable, analysis-ready datasets and actionable business insights for HR, marketing, and industrial operations. BE in Computer Science and Engineering.

TECHNICAL SKILLS

Data Analytics & Visualization: Power BI, MS Excel, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Data Gathering, Data Cleansing, Normalization, Exploratory Data Analysis, KPI Reporting

SQL & Databases: SQL (Joins, Views, CTEs, Window Functions), SQL Server, PostgreSQL, MySQL, MongoDB, DBMS Fundamentals, Relational Database Design, Data Warehousing

Data Engineering: ETL Processes, Data Pipelines, Document Ingestion, Data Integration, Python Automation

Programming Languages: Python, SQL, C, Java, JavaScript, HTML, CSS

GenAI & Machine Learning: RAG, LangChain, Ollama, Hugging Face Transformers, OpenAI API, Prompt Engineering, Embeddings, Semantic Search, FAISS, ChromaDB, TensorFlow, Keras, Scikit-learn

Backend & Tools: FastAPI, Flask, REST APIs, Docker, Podman, Git, GitHub, Google Cloud Platform (GCP), Jupyter, VS Code

PROFESSIONAL EXPERIENCE

ASM Technologies – Software Engineer **Aug 2025 – Present**

- Built an end-to-end Retrieval-Augmented Generation (RAG) assistant for industrial failure analysis, supporting troubleshooting, log analysis, and automated FA report generation.
- Designed document ingestion and ETL-style pipelines (parsing, cleansing, chunking, embedding) loading technical documents into ChromaDB for semantic retrieval using LangChain and Ollama LLMs.
- Developed REST APIs with FastAPI and Flask, integrating PostgreSQL and MongoDB for structured data storage, chat history, and application workflows.
- Applied prompt engineering, retrieval optimization, and context management to improve response quality and reduce hallucinations.
- Analyzed log data to identify failure patterns and root causes, generating citation-backed failure analysis reports for engineers.

AES Technologies – Software Engineering Intern (HRIQ HRMS Project) **Oct 2023 – May 2024**

- Contributed to full-stack development of an HRMS platform for employee management, owning the Reimbursement feature.
- Analyzed employee and reimbursement data using SQL and .NET to identify trends and improve processing efficiency.
- Collaborated in a cross-functional team of 6 using jQuery, C#, and SQL Server to deliver a scalable, modular platform, reducing feature deployment time by 20%.

PROJECTS

HR & Marketing Analytics – Solo Project (Python, SQL Server, Power BI, Pandas, NumPy)

- Analyzed HR data on employee performance and retention to derive actionable insights for workforce optimization.
- Analyzed retail shop data to understand customer behavior and recommend improvements to business and marketing strategy.
- Gathered, cleansed, and normalized raw data using Python (Pandas, NumPy) and loaded it into SQL Server for warehousing and reporting.
- Built interactive Power BI dashboards to communicate trends and key metrics.

Automotive Failure Detection and Recovery Recommendation

- Built a hybrid LLM + rule-based system that classified automotive failures with ~90% accuracy across 4 OEM datasets, generated recovery recommendations, and delivered a web UI.
- Compared GPT-3, Llama 3, Claude, Falcon, and Bard outputs to select the best-performing model. Tech: Python, React, Node.js.

Bearing Fault Classification (Published Paper)

- Developed deep learning and machine learning models to predict bearing faults and remaining lifetime from sensor data. Tech: Python, CNN, ANN, SVM, LightGBM, TensorFlow.

EDUCATION

PSG College of Engineering, Coimbatore – B.E. Computer Science and Engineering (CGPA: 8.0) **Nov 2021 – Apr 2025**

CS Academy, Erode – Class XII: 94% (2021), Class X: 92% (2019) **May 2019 – May 2021**

ACHIEVEMENTS & LEADERSHIP

- Represented ASM Technologies at CII INNOVERGE 2025; delivered a technical talk on "Building a RAG Chatbot using NVIDIA NIM".
- Secretary, Entrepreneur's Club – led event organisation and speaker sessions.